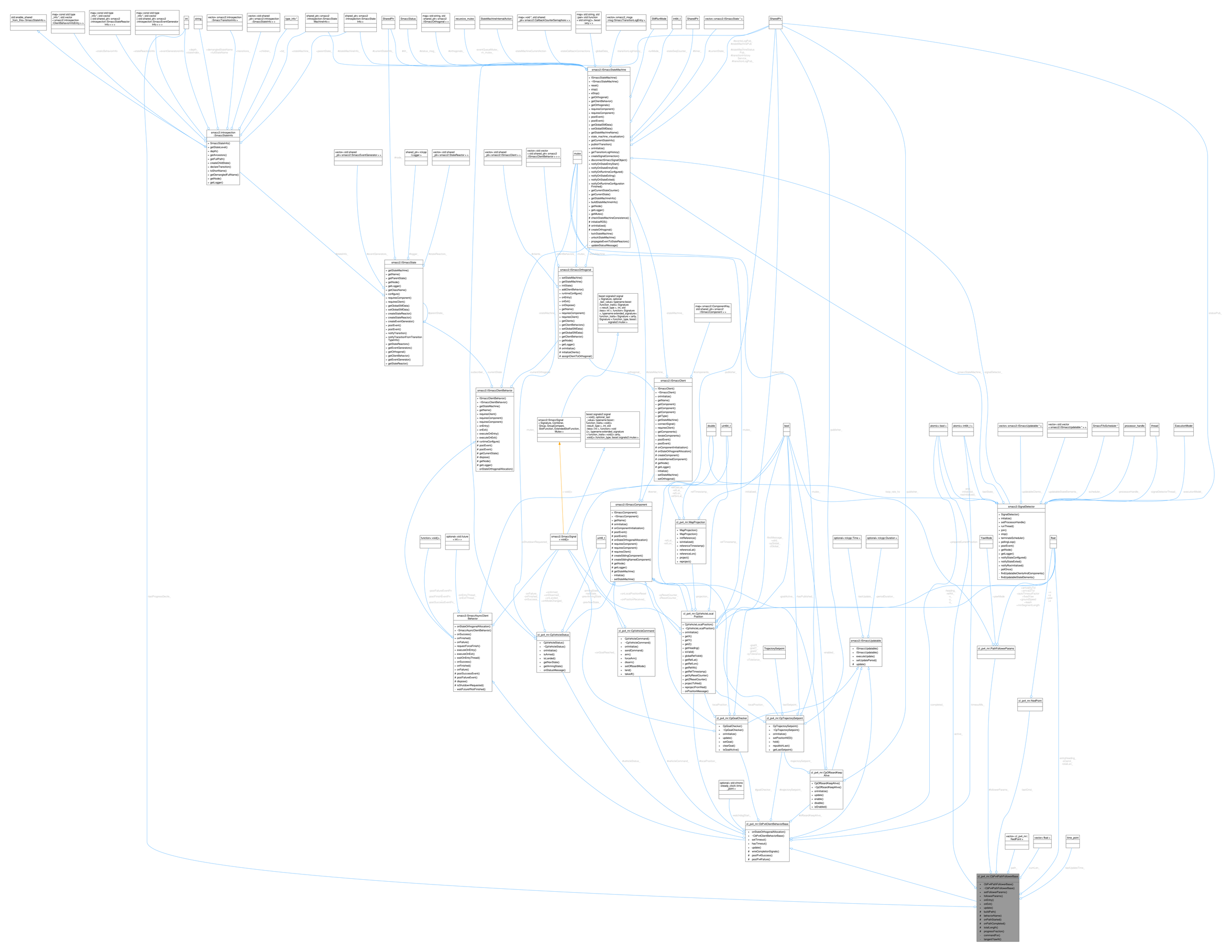

std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
int
string
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
shared_ptr< smacc2::introspection::SmaccStateInfo >
SharedPtr
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
recursive_mutex
StateMachineInternalAction
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
SharedPtr
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+depth_
+stateIndex_
+demangledStateName
+fullStateName
+transitions_
+children_
+tid_
+stateMachine_
+parentState_
#stateMachineInfo_
#currentStateInfo_
#nh_
#status_msg_
#orthogonals_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#eventsLogPub_
#stateMachinePub
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
mutex
#node_
#stateInfo_
#eventGenerators_
#logger_
#stateReactors_
#clients_
-clientBehaviors_
-mutex_
-stateMachine
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
-stateMachine_
-stateMachine_
-subscriber_
-currentState
-currentOrthogonal
-orthogonal_
#stateMachine_
#components_
-publisher_
-subscriber_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void ()>, typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
-mutex_
-mutex_
-publisher_
double
uint64_t
bool
< void()>
#owner_
-refCosLat_
-refLat_
-refLon_
-refSinLat_
-refTimestamp_
-initialized_
-mutex_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
cl_px4_mr::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
atomic< bool >
atomic< int64_t >
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-smaccStateMachine_
-signalDetector_
-statusPub_
-publisher_
-mutex_
-loop_rate_hz
-publisher_
-end_
-initialized_
-rosInitialized_
-lastState_
-updatableClients_
-updatableStateElements_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
function< void()>
optional< std::future< int > >
smacc2::SmaccSignal< void()>
uint8_t
-isShutdownRequested_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
-onEntryThread_
-onExitThread_
-onFailure_
-onFinished_
-onSuccess_
+onArmed_
+onDisarmed_
+onLanded_
+onModeChanged_
-armingState_
-navState_
-prevArmingState
-prevNavState_
+onLocalPositionReset
+onPositionReceived_
-xyResetCounter_
-zResetCounter_
-projection_
-refLat_
-refLon_
-refTimestamp_
-firstMessage_
-valid_
-xyGlobal_
-zGlobal_
-goalActive_
-hasPublished_
-lastProgressDecile_
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
cl_px4_mr::CpVehicleStatus
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
cl_px4_mr::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
+onGoalReached_
cl_px4_mr::CpVehicleLocalPosition
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
-goalX_
-goalY_
-goalZ_
-xyTolerance
-zTolerance_
TrajectorySetpoint
-enabled_
-localPosition_
-localPosition_
-lastSetpoint_
cl_px4_mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
cl_px4_mr::CpTrajectorySetpoint
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
-trajectorySetpoint_
cl_px4_mr::CpOffboardKeepAlive
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
#vehicleStatus_
#vehicleCommand_
#localPosition_
#goalChecker_
#trajectorySetpoint_
#offboardKeepAlive_
optional< std::chrono::steady_clock::time_point >
-watchdogStart_
cl_px4_mr::CbPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~CbPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
optional< rclcpp::Time >
optional< rclcpp::Duration >
-lastUpdate_
-periodDuration_
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
YawMode
float
+prependCurrentPosition
-heading_
-refAlt_
-x_
-y_
-z_
+yawMode
+arrivalXyTol
+arrivalZTol
+autoTimeoutFactor
+fixedYaw
+groundSpeed
+leash
+minSegmentLength
+x
+y
+yaw
+z
cl_px4_mr::PathFollowerParams
cl_px4_mr::NedPoint
-completed_
-timeoutMs_
-active_
-entryHeading_
-sCarrot_
-totalLen_
#followerParams_
-lastCmd_
vector< cl_px4_mr::NedPoint >
vector< float >
time_point
-path_
-cumLen_
-lastUpdateTime_
cl_px4_mr::CbPx4PathFollowerBase
+ CbPx4PathFollowerBase()
+ ~CbPx4PathFollowerBase()
+ setFollowerParams()
+ followerParams()
+ onEntry()
+ onExit()
+ update()
# buildPath()
# behaviorName()
# onPathStarted()
# onPathCompleted()
# totalLength()
# progressFraction()
- commandFor()
- tangentYawAt()